

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Olawale Sogbesan
Program/Batch	B082-Networkwalks
Date	20 August 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM4 (TheHarvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. Microsoft.com 3. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1), [Microsoft.com](https://www.microsoft.com) using theHarvester (W2-PM4) and scanning my own local network with Zenmap (W2-PM5). Two modules cover the footprinting phase and the last covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
theHarvester	Gathering information of emails, sub-domains, hosts, employee names, open ports and banners from different public sources
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I did reconnaissance on networkwalks.com domain using six Kali Linux tools **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f** and **DNSRecon**. Each tool was used to collect a different kind of information about the target.

First, I used WHOIS to obtain publicly available domain registration information and determine the domain's name servers. The results gave insights into the domain registration and hosting infrastructure.

I then used WhatWeb to discover what technologies are used by the website. The results identified WordPress 7.0.4 and WP Download Manager 3.3.58 and other information exposed by the website.

I did a nslookup and got the ip address of the domain name. This time it returned 192.232.216.135.

I used **Curl** with the -I option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint /wp-json/.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Lastly, I used DNSRecon to enumerate DNS records. The results gave information like name servers, mail servers, SPF / TXT records, service records and DNS software information.

4.2 FOOTPRINTING & RECONNAISSANCE ATTACKS WITH theHARVESTER

For the second activity, I used theHarvester to perform [passive] footprinting on the target microsoft.com. The practical required me to gather information of emails, sub-domains, hosts, employee names, open ports and banners from different public sources like search engines, PGP key servers and SHODAN computer database

4.3 Network Scanning with Zenmap

For the third activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified two live hosts:

- 192.168.4.1
- 192.168.18.5

The example results also included four MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● Medium
6	Massive subdomain exposure	9,484 hosts enumerated (e.g., *.copilotstudio.microsoft.com, *.prod.api.copilotforfinance.microsoft.com, internal-looking names like redmond.corp.microsoft.com, sys-wingroup.ntdev.corp.microsoft.com)	Attackers can map the external attack surface, discover forgotten or pre-production hosts, and prioritize targets for further scanning (port scans, vuln scans, subdomain takeover attempts)	● Critical
7	Environment/staging host disclosure	Many -ppe, -dev, -uat, -test, -sdf, -int labeled hosts (e.g., aicc-dev.microsoft.com, askiam-uat.microsoft.com, admin-sdf.microsoft.com)	Pre-production systems are frequently less hardened, more likely to have default credentials, verbose errors, or outdated patches — attractive targets for initial compromise	● Critical

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
8	Public IP address disclosure	122 IPs identified across multiple ranges (Azure, Akamai, Cloudflare-fronted, and direct Microsoft-owned blocks like 13.107.x.x, 150.171.x.x, 40.x.x.x)	Enables direct network-layer reconnaissance (port scanning, service fingerprinting, geolocation of infrastructure) bypassing any CDN/WAF fronting	● Medium
9	Employee email addresses harvested	3 addresses found (dotnet-docker-bot@microsoft.com, opencode@microsoft.com, secure@microsoft.com)	Limited in volume, but usable for phishing pretexting or as a starting point for further email-format guessing (first.last@microsoft.com patterns)	● Low
10	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

- 1. Review publicly exposed technology information**
Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
- 2. Keep software updated**
CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
- 3. Review HTTP headers**
HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.

4. **Review DNS records regularly**
DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used seven Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

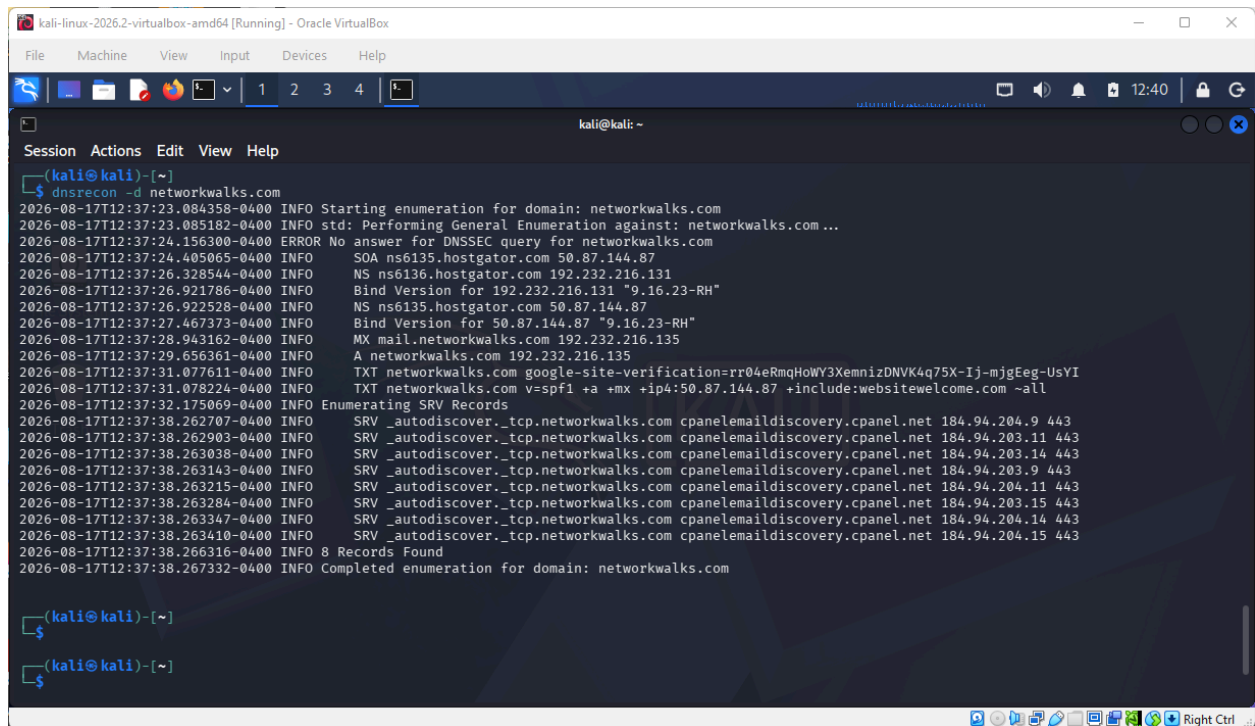
In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

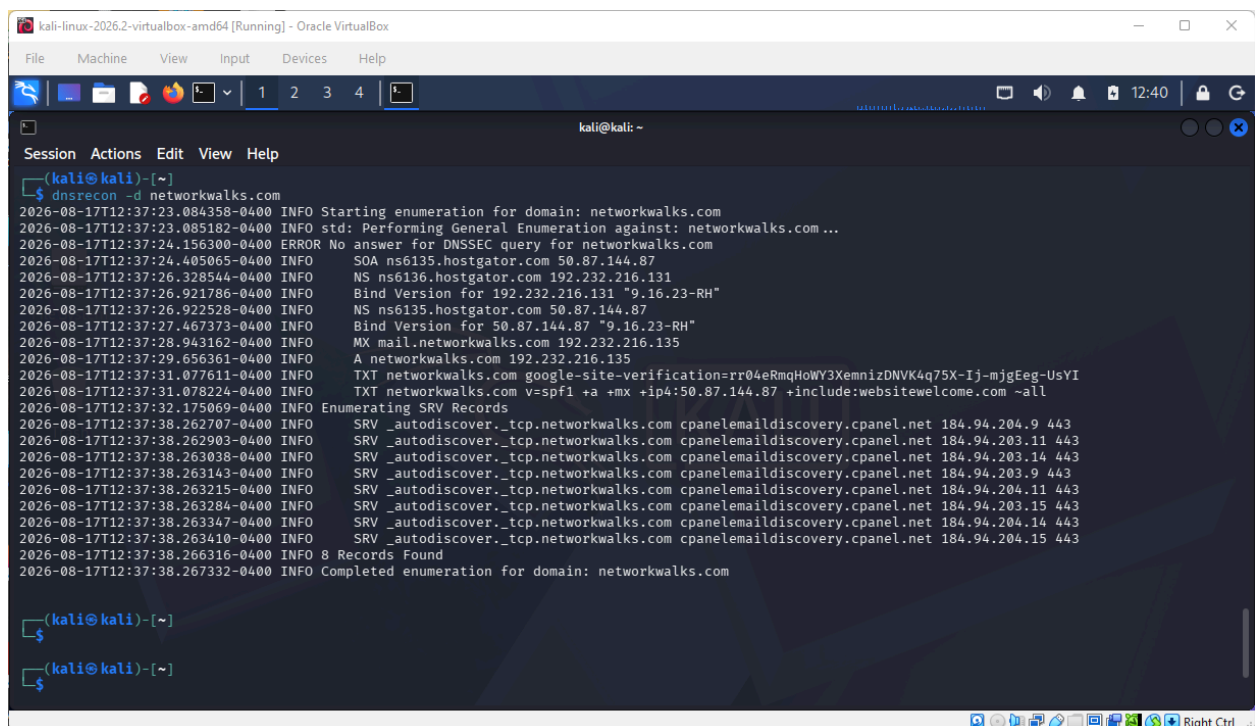
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected



```
kali@kali: ~  
$ dnsrecon -d networkwalks.com  
2026-08-17T12:37:23.084358-0400 INFO Starting enumeration for domain: networkwalks.com  
2026-08-17T12:37:23.085182-0400 INFO std: Performing General Enumeration against: networkwalks.com ...  
2026-08-17T12:37:24.156300-0400 ERROR No answer for DNSSEC query for networkwalks.com  
2026-08-17T12:37:24.405065-0400 INFO SOA ns6135.hostgator.com 50.87.144.87  
2026-08-17T12:37:26.328544-0400 INFO NS ns6136.hostgator.com 192.232.216.131  
2026-08-17T12:37:26.921786-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"  
2026-08-17T12:37:26.922528-0400 INFO NS ns6135.hostgator.com 50.87.144.87  
2026-08-17T12:37:27.467373-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"  
2026-08-17T12:37:28.943162-0400 INFO MX mail.networkwalks.com 192.232.216.135  
2026-08-17T12:37:29.656361-0400 INFO A networkwalks.com 192.232.216.135  
2026-08-17T12:37:31.077611-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI  
2026-08-17T12:37:31.078224-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all  
2026-08-17T12:37:32.175069-0400 INFO Enumerating SRV Records  
2026-08-17T12:37:38.262707-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443  
2026-08-17T12:37:38.262903-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443  
2026-08-17T12:37:38.263038-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443  
2026-08-17T12:37:38.263143-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443  
2026-08-17T12:37:38.263215-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443  
2026-08-17T12:37:38.263284-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443  
2026-08-17T12:37:38.263347-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443  
2026-08-17T12:37:38.263410-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443  
2026-08-17T12:37:38.266316-0400 INFO 8 Records Found  
2026-08-17T12:37:38.267332-0400 INFO Completed enumeration for domain: networkwalks.com  
  
kali@kali: ~  
$
```



```
kali@kali: ~  
$ dnsrecon -d networkwalks.com  
2026-08-17T12:37:23.084358-0400 INFO Starting enumeration for domain: networkwalks.com  
2026-08-17T12:37:23.085182-0400 INFO std: Performing General Enumeration against: networkwalks.com ...  
2026-08-17T12:37:24.156300-0400 ERROR No answer for DNSSEC query for networkwalks.com  
2026-08-17T12:37:24.405065-0400 INFO SOA ns6135.hostgator.com 50.87.144.87  
2026-08-17T12:37:26.328544-0400 INFO NS ns6136.hostgator.com 192.232.216.131  
2026-08-17T12:37:26.921786-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"  
2026-08-17T12:37:26.922528-0400 INFO NS ns6135.hostgator.com 50.87.144.87  
2026-08-17T12:37:27.467373-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"  
2026-08-17T12:37:28.943162-0400 INFO MX mail.networkwalks.com 192.232.216.135  
2026-08-17T12:37:29.656361-0400 INFO A networkwalks.com 192.232.216.135  
2026-08-17T12:37:31.077611-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI  
2026-08-17T12:37:31.078224-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all  
2026-08-17T12:37:32.175069-0400 INFO Enumerating SRV Records  
2026-08-17T12:37:38.262707-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443  
2026-08-17T12:37:38.262903-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443  
2026-08-17T12:37:38.263038-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443  
2026-08-17T12:37:38.263143-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443  
2026-08-17T12:37:38.263215-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443  
2026-08-17T12:37:38.263284-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443  
2026-08-17T12:37:38.263347-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443  
2026-08-17T12:37:38.263410-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443  
2026-08-17T12:37:38.266316-0400 INFO 8 Records Found  
2026-08-17T12:37:38.267332-0400 INFO Completed enumeration for domain: networkwalks.com  
  
kali@kali: ~  
$
```

```
(kali@kali)-[~]
$ wafw00f networkwalks.com
File System
{ WOOF! }
404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.0.4], Cookies[_wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=505116&pvtVar2=76&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.0.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=505116&pvtVar2=76&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.4]

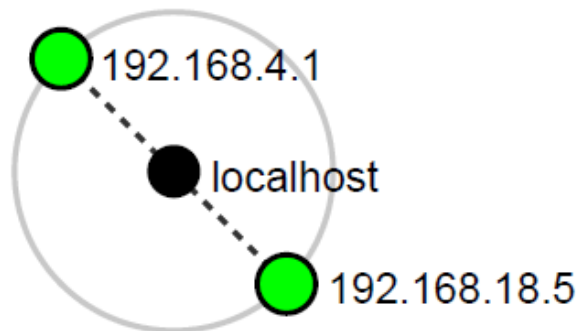
(kali@kali)-[~]
$
```



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~
Session Actions Edit View Help
(kali@kali)~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-08-17T15:38:57Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
```



-End-



Author

Olawale Sogbesan

Cybersecurity Professional B082

LinkedIn: <https://www.linkedin.com/in/olawale-sogbesan-611734160/>



Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub